

Italian Road Risk Analysis



Boolean Capstone Project

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Data Analytics

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Problem Statement & Objectives

Total accident counts often hide the real danger. Large hubs (e.g. Milan) always lead the rankings simply due to their size, not necessarily due to poorer safety.

To discover the "true" risk, we must compare accidents to other metrics, in order to identify statistically significant urban hotspots.

Main Objective: Gain insights into accident rates across Italian municipalities to identify high-risk areas where major infrastructure or safety interventions are needed.

Scope: The analysis covers a 10-year historical period (2014–2024), utilizing official data from ISTAT.



Process & methodology - part 1: Python

Using Python (on VS Code as an IDE), we performed a process of Data fetching, Data cleaning, Feature engineering and Exploratory Data Analysis.

The main libraries used are Selenium for data scraping, Pandas and Matplotlib for cleaning and feature engineering, and Seaborn for charting. Also, we used Scikit-Learn for the implementation of clustering and data modeling (Logistic Regression).

01 - Data Fetching

Data acquisition from ISTAT (via API), Situas (via scraping), and ACI. Performed initial data cleaning.

02 - Data Cleaning/Feature Engineering

Data cleaning and preparation of the main datasets for EDA, incorporating key metrics (e.g., Accident rates).

03 - Exploratory Data Analysis

Analysis of temporal trends and charting of top municipalities by various metrics (total accidents, accident rates, etc.).

04 - Machine Learning Models

Implementation of clustering models (K-Means and DBSCAN) and a Logistic Regression model.



Process & methodology - part 1: Power BI

On Power BI, we started by uploading the datasets generated in Python. We checked the consistency of the columns and created the star schema. Lastly, we built the dashboard, implementing the necessary measures along the way.

01 - Uploading .csv and general check

Uploaded the various .csv files created, checking data types and modifying them if needed.

02 - Data model

Created a star schema model, establishing the required relationships between dimensional and fact tables.

03 - Dashboard

Created the dashboard, adding measures along the way when needed. We also decided (for better visibility) to filter for municipalities with 1,000 inhabitants or more.

Insights & Conclusions

In this analysis, we gained some major insights:

- The total number of accidents remained relatively constant over the years
- During the COVID-19 pandemic, while total accidents went down, the fatality rate and severity of crashes rose.
- If we look at absolute numbers, big hubs like Milan or Rome are at the top of the chart; however, when correlating accidents with population or surface area, several minor hubs (especially tourist areas) become notable.
- At a regional granularity, the main area for investment is Liguria.

Thank you for your attention

